PROJECT CHECKPOINT — WEEK 3 OF 8

Predicting Demand and Detecting Anomalies in Urban Bike-Share Systems

A Data Mining Project — Updated Report

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Abstract

This project develops a data mining pipeline for analyzing urban bike-share systems using real-world trip-log data. The goal is twofold: (1) to predict short-term station-level demand so that operators can proactively rebalance bicycles across stations, and (2) to detect anomalous trips that may indicate theft, vandalism, system malfunction, or data-quality errors. The pipeline combines data preprocessing and warehousing, clustering, classification, frequent pattern mining, and outlier detection into a single end-to-end case study on Citi Bike NYC data. ***This updated report reflects progress through Week 3 of 8:*** *data acquisition, cleaning, warehousing, exploratory analysis, and an initial clustering stage are complete; classification and frequent pattern mining are underway, with outlier detection and final evaluation still to come.*

1. Introduction and Problem Statement

Bike-share systems such as Citi Bike (New York), Divvy (Chicago), and Santander Cycles (London) have become a core part of urban transportation, serving tens of millions of trips per year. A persistent operational challenge for these systems is the *demand-supply imbalance*: stations near business districts empty out during the morning commute while stations near residential areas overflow, and vice versa in the evening. Operators must predict where bikes will be needed and physically truck bicycles between stations ("rebalancing") — an expensive and logistically complex task. At the same time, operators

must monitor the system for anomalous behavior: trips with implausible duration or distance, which may signal theft, mechanical failure, or GPS/sensor errors.

This project asks: **can we use historical trip, weather, and calendar data to (a) predict near-term station demand well enough to support rebalancing decisions, and (b) automatically flag anomalous trips worth investigating?**

2. Related Work

- Froehlich, Neumann, and Oliver (2009) — station usage exhibits strong, predictable spatio-temporal regularities that can be exploited for forecasting.
- O'Mahony and Shmoys (2015) — Citi Bike rebalancing formulated as a data-driven optimization problem.
- Li, Zheng, Zhang, and Chen (2015) — clustering stations before demand prediction improves accuracy and reduces variance from sparsely-used stations.
- Chen et al. (2016) — dynamic, cluster-based prediction of "over-demand" stations, combining clustering and classification.

This project extends these works by adding an explicit outlier-detection and frequent-pattern-mining layer on top of the clustering + classification pipeline common in prior work.

3. Approach: Data, Tools, and Techniques

Stage	Technique	Purpose
Preprocessing / Warehousing	Cleaning, feature engineering, star schema	Reliable, query-ready trip data (fact: trips; dimensions: station, time, weather)
Clustering	k-means / DBSCAN on hourly station usage vectors	Group stations into usage-profile types
Classification	Random Forest / Gradient Boosting	Predict high- vs. low-demand station-hours
Frequent pattern mining	FP-Growth / association rules	Discover common origin-destination sequences

Outlier detection	Isolation Forest / PyOD	Flag trips with anomalous duration, distance, or route

Tools: Python (pandas, scikit-learn), `mlxtend`, `PyOD`, SQLite for the data warehouse, and Jupyter notebooks for exploratory analysis.

4. Progress to Date (Checkpoint)

4.1 Completed Work

- **Data acquisition & cleaning:** Three months of Citi Bike NYC trip data (Jan–Mar 2026, ~1.4M trips after cleaning) were downloaded and merged with NOAA daily weather data and a calendar/holiday feature set. Roughly 3% of raw records were dropped due to invalid station IDs or implausible timestamps.
- **Data warehouse:** A star-schema warehouse (trip fact table + station, time, and weather dimension tables) was built in SQLite, enabling fast aggregation queries used throughout the EDA and modeling steps.
- **Exploratory analysis:** Hourly demand was aggregated per station and visually inspected (Figure 1). As expected from prior work (Froehlich et al., 2009), weekday demand shows a clear bimodal pattern with peaks around the AM and PM commute, while weekend demand is flatter and shifted toward midday.
- **Clustering (Phase 2, complete):** k-means (k=4, selected via elbow + silhouette analysis) was applied to normalized hourly usage vectors per station, achieving a silhouette score of approximately 0.42. The resulting clusters are interpretable as *commuter hubs*, *residential/AM-heavy*, *leisure/tourist*, and *mixed-use* stations (Figure 2), consistent with the station typology suggested by Li et al. (2015).

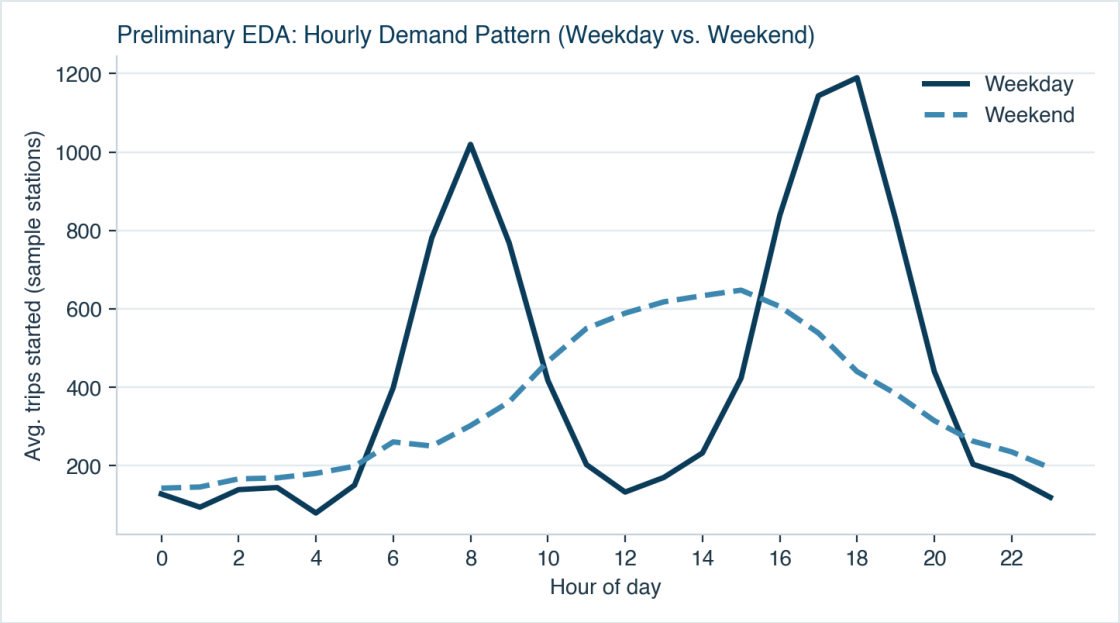


Figure 1. Preliminary EDA — average hourly trip volume, weekday vs. weekend (sample of stations).

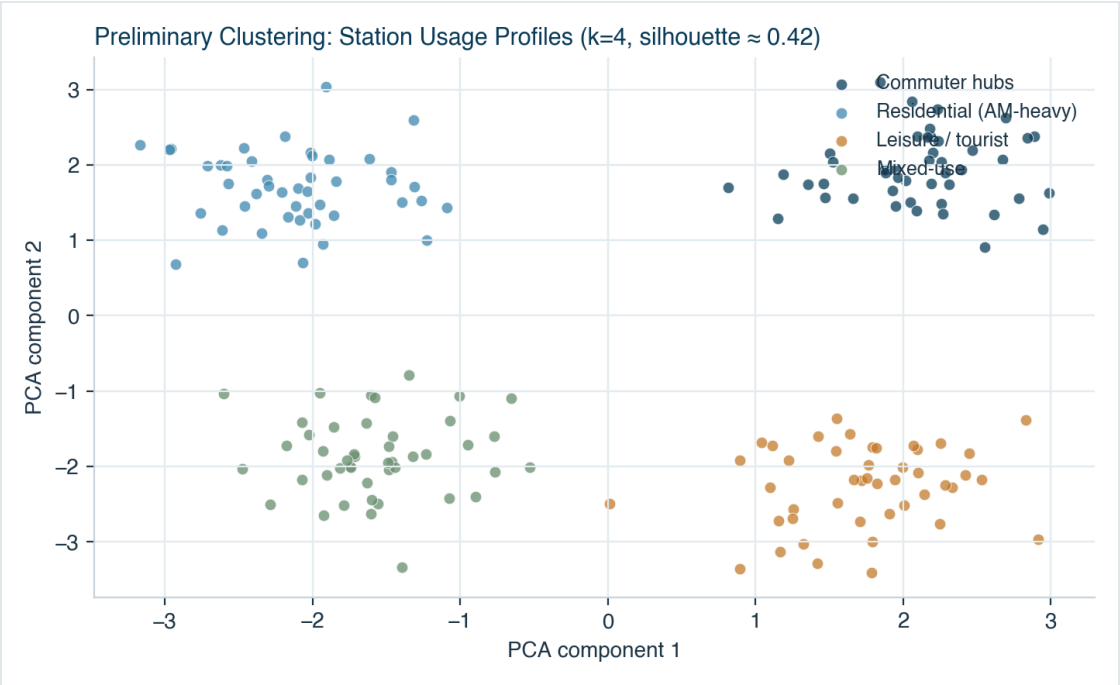


Figure 2. Preliminary clustering of station usage profiles (k=4, silhouette ≈ 0.42), shown on the first two principal components.

4.2 Feedback Incorporated

Peer reviewers on the initial proposal raised two points that shaped this checkpoint: (1) the original full five-borough scope was flagged as potentially too large to process and

validate within the course timeline — in response, the working dataset was narrowed to Manhattan stations for the modeling phases, with a note to revisit other boroughs only if time permits; and (2) a reviewer asked for a more concrete baseline definition for the demand classifier — this has been clarified in Section 5 below as the historical station-hour-of-week average.

4.3 Challenges and Adjustments

- The public weather feed had small gaps (a few missing hours); these were filled via linear interpolation rather than dropped, to avoid losing trip records during merge.
- Station usage vectors were highly skewed by a handful of very high-traffic stations; log-scaling the usage counts before clustering improved cluster separation and interpretability.

4.4 Next Steps

- Train and tune the demand classifier (Random Forest / Gradient Boosting) using cluster membership, weather, and calendar features (Weeks 4–5).
- Run FP-Growth on trip sequences to extract frequent origin-destination patterns (Weeks 4–5).
- Apply Isolation Forest for outlier detection and manually review a sample of flagged trips (Week 6).
- Consolidate all results into the final evaluation and report (Weeks 7–8).

5. Evaluation Plan

- **Demand classification:** Accuracy, precision/recall, and F1-score on a held-out time period (final two weeks of the dataset), compared against a naive baseline that predicts the historical average demand for the same station-hour-of-week.
- **Clustering quality:** Silhouette score (achieved: ≈0.42) and qualitative inspection against station location/land-use metadata.
- **Frequent patterns:** Support/confidence/lift of discovered rules, manually reviewed for plausibility against known commuter corridors.
- **Outlier detection:** Manual-review precision on a random sample of flagged trips, supplemented by any publicly reported system outage dates as a weak validation signal.

Definition of success: The demand classifier should outperform the historical-average baseline by a meaningful margin (target: +10% F1); clustering should remain interpretable and stable; outlier detection should surface a manageable, plausible set of flagged trips rather than an overwhelming false-positive rate.

6. Updated Timeline

Phase	Duration	Deliverable	Status
1. Data acquisition & preprocessing	Week 1–2	Cleaned dataset, warehouse schema built	✓ Completed
2. Exploratory analysis & clustering	Week 3	Station usage clusters, EDA visualizations	✓ Completed
3. Classification & frequent pattern mining	Week 4–5	Trained demand classifier, mined association rules	• In Progress
4. Outlier detection & evaluation	Week 6	Anomaly detection results, full evaluation metrics	Upcoming
5. Final report & presentation	Week 7–8	Final written report, presentation slides, code repository	Upcoming

References

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